

Concept Plans and Spatial Organization Diagrams

How the park is organised — the crescent, the radial alleys, and the rooms between them.

03

OF 12 UPLOAD SLOTS

144 m

CRESCENT ARC LENGTH

15,000 m²

SITE AREA

34.1%

SITE-WIDE MEAN SHADE

131

TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL
wasim437.github.io/AI_SAFA/

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01 WHAT IS IN THIS FILE

Phase4 Concept Development Report	DRAWING
Circulation	DRAWING
Facilities	DRAWING
Fig04 site comfort map	FIGURE
Fig10 masterplan	PLAN

02 THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

Produced by **CODE** [src/plan.py](#) [src/drawings.py](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

03

OF 12

01 THE TEN PHASES

P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P5 Masterplan Development

THIS DOCUMENT

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P4 Concept Development

THIS DOCUMENT

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Plan

src

Drawings

src

03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 03 OF 12 · DESIGN NARRATIVE & CONCEPT

Design Narrative & Concept

Falaj Al Safa — an arc of shade for a park that is currently too hot to use

Of the 4,402 daylight hours in a Dubai year, only **44.5%** are comfortable to stand in on the open site today. This proposal raises that to **64.6%** — a gain of 20.1 percentage points, or about 884 additional usable hours a year.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 05 August 2026 · every quantity regenerated by `python run_analysis.py`

1. The problem, stated as a number

A neighbourhood park nobody can stand in is not a park. Al Safa 2's difficulty is not layout, planting or maintenance — it is thermal. The site is flat and largely exposed, at 25°N, where the summer sun is nearly overhead and the winter sun is low enough to slide underneath most shade.

The analysis behind this submission reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, verified back against those normals to within 0.39 °C. Across the 4,402 daylight hours in that year, the open site is comfortable for 44.5% of them. That is the number the design exists to move.

2. The concept

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

Element	What it is	Description
Al Hilal	the Crescent Canopy	18 m gridshell at 4.5 m over a 7 m walk, with a 3 m southern louvre
Al Falaj	the water channel	0.9 m wide, on the canopy's drip line so it is shaded all day and evaporates less
Al Nakhil	the Oasis Basin	a sunken palm court held in the crescent's concave side
Al Sikkak	the alleys	radial — each one a radius of the same arc
Al Madar	the perimeter loop	a tree-shaded running and walking circuit
Al Kathib	the dune berm	planted earth against the roads — noise, glare and heat

The six named elements. All are generated from a single arc definition in `src/plan.py`.

3. Why an arc — the part that was solved, not styled

A straight canopy presents one orientation. When a sun angle defeats it, it defeats the whole length at once, and the walk has no shade anywhere along it. An arc changes heading continuously, so some segment is always angled well.

The depth of the bow was not chosen for appearance. It was swept against the 8,760-hour solar model at a fixed section, over all 4,402 daylight hours of the year. The criterion was the final column — the hours in which the route offers nowhere at all to stand.

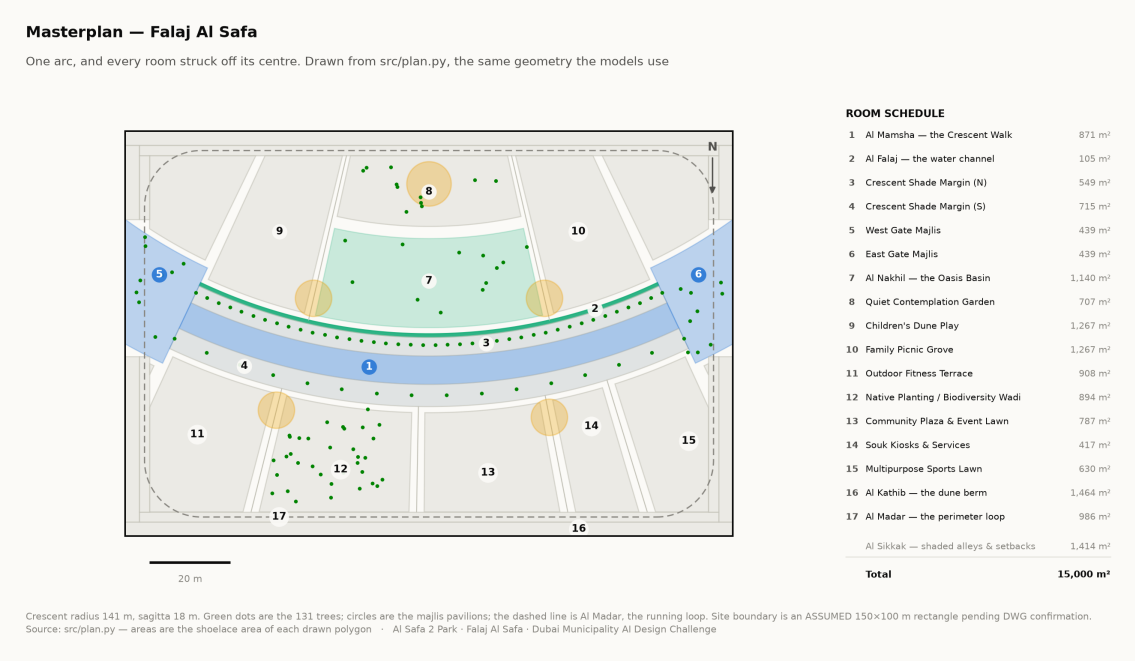
Plan form	Mean cover	Worst month	Hours with no shade anywhere
Straight east–west bar	87.4%	68.7%	330
Sine meander (superseded)	85.0%	73.1%	63
Arc, sagitta 10 m	87.1%	70.2%	116
Arc, sagitta 14 m	86.6%	71.3%	62
Arc, sagitta 18 m	85.9%	72.1%	52
Arc, sagitta 22 m	84.9%	72.3%	61
Closed elliptical loop	79.1%	69.8%	89

Adopted row highlighted. Source: the sweep recorded in src/config.py, measured against the same solar model used throughout this submission.

Read the first column carefully. The straight bar has the *highest* mean coverage of anything tested — an east–west canopy is close to the optimum orientation for 25°N, and curving it costs about a point. The crescent is not claimed to shade more ground on average. It is claimed to remove the hours in which the route offers nowhere at all to stand, and it removes six sevenths of them. That is the trade, stated in the direction that is not flattering.

The closed loop loses on every measure, for the same reason the deep bows do: it forces half its length to run north–south, and a canopy over a north–south route can only work when the sun is low in the east or west. The loop survives in the scheme as Al Madar, an *unshaded* running circuit — because that is what a circuit is actually for.

The arc bows convex south. The solar model is indifferent to that sign, so the choice was spatial: bowing south puts the structure between the sun and the hollow it wraps, making the concave side the park's cool pocket instead of a south-facing bowl. Every room people are asked to linger in sits in that hollow.



Masterplan. Every area is the shoelace area of the drawn polygon, which is why the schedule closes on 15,000 m² without anyone reconciling a spreadsheet. Technical drawing — to scale.

4. What the design achieves

Measure	Today	As designed
Comfortable daylight hours	44.5%	64.6%
Peak heat index	56.8 °C	48.7 °C
Mean heat-index reduction under canopy	—	7.13 °C
Crescent Walk shaded (canopy and louvre)	—	87.3%
Site-wide mean shade	—	34.1%
Trees	—	131

Site-wide mean shade is modest at 34.1%, and that is a position, not an oversight. This scheme makes a few places genuinely excellent rather than the whole site marginally better. A park with even shade everywhere and no continuously comfortable route would score higher on that one statistic and be worse to use.

5. Two corrections this submission makes to itself

A withdrawn shade claim. An earlier version claimed 99.2% annual shade on a flat 9 m canopy over a 9 m walkway. It does not survive a geometric check — when the sun is low and to the south, the shadow slides clean off the path — and it was withdrawn. The section was then re-solved: a narrower walk, a wider overhang, a lower plane and a deep southern louvre. It now measures 87.3%, and that is a number that can be checked.

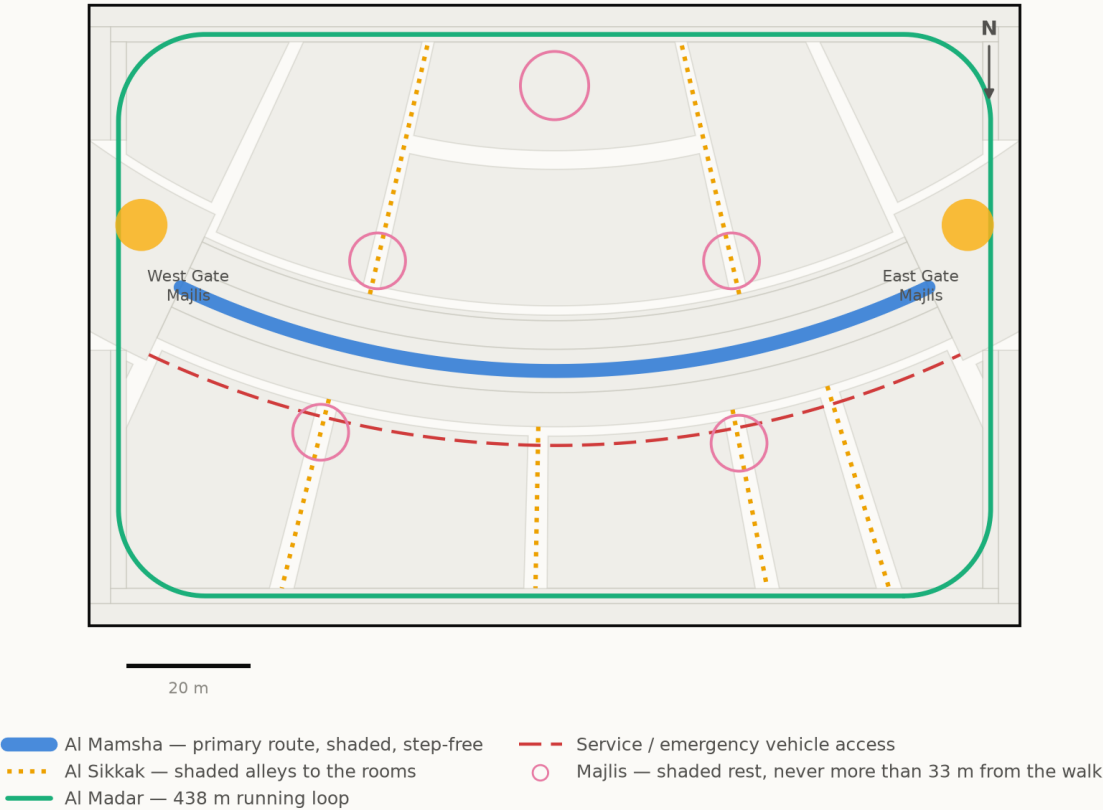
Three withdrawn images. Three visuals presented invented data as measurement: a thermal comfort analysis attributed to CFD when no CFD was run; a satellite NDVI analysis whose raster was noise; and a parametric canopy described as solar-optimised when no objective was ever defined. All three were withdrawn on the same principle as the 99.2%: anything that cannot survive being checked should not be in front of a juror.

Circulation

Every route through the park, and the step-free claim it supports

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: [src/plan.py](#) — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/drawings.py](#) [src/plan.py](#)

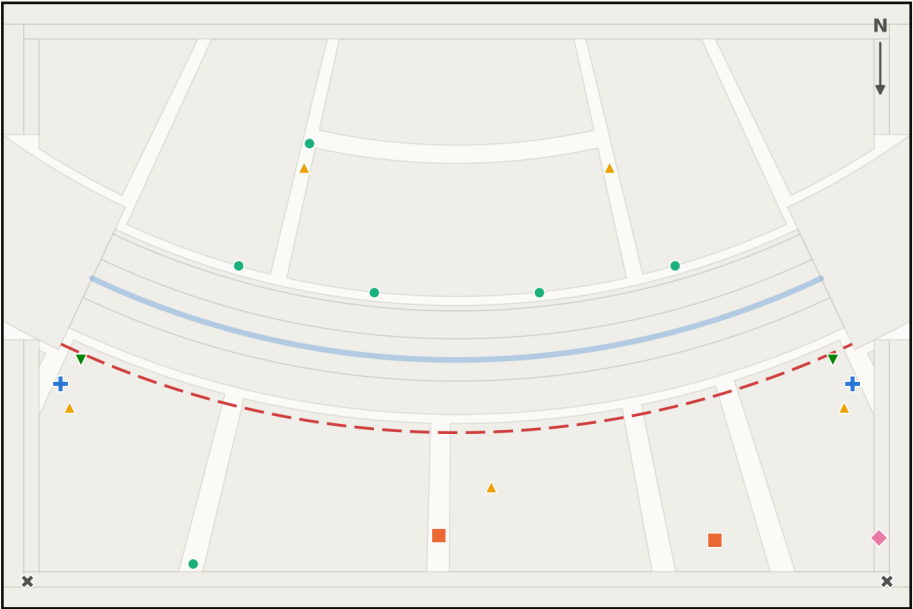
Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with `python run_analysis.py`

Facilities

The 20 facilities the brief requires, placed and counted

Commercial & service facilities

Where the park earns its running costs, and where it is serviced



- Service / emergency vehicle access
- Commercial — souk kiosks and café (2)
- Public restrooms (2)
- Drinking fountain (6)
- Waste & recycling station (5)
- Service & maintenance store (1)
- Bicycle parking (2)
- Drop-off / pick-up bay (2)

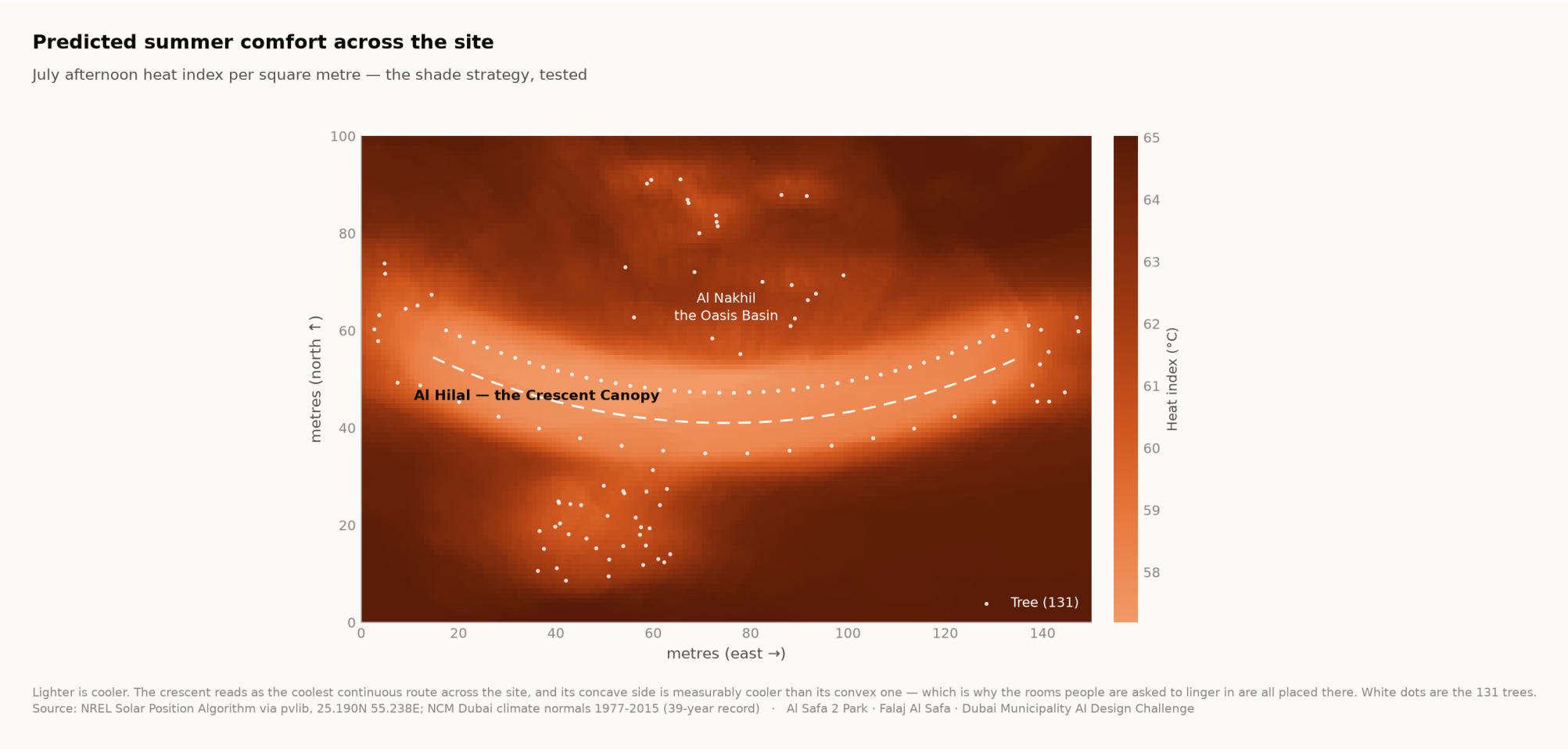
20 facilities. Commercial floor sits on the convex south face (417 m2, 2.8% of the site), facing the plaza and the event lawn so that evening trade and programming reinforce each other — and away from the concave hollow, which is kept quiet. Every facility is reachable from the dashed service route without a vehicle entering the walk.
Source: [src/plan.py](#) — facility positions are arc coordinates, so they move with the crescent · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/plan.py](#) [src/drawings.py](#)

Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with `python run_analysis.py`

Fig04 site comfort map

July afternoon heat index for every square metre of the site



DRAWN BY [src/solar.py](#) [src/figures.py](#) READ FROM [data/processed/spatial_grid_comfort.csv](#)

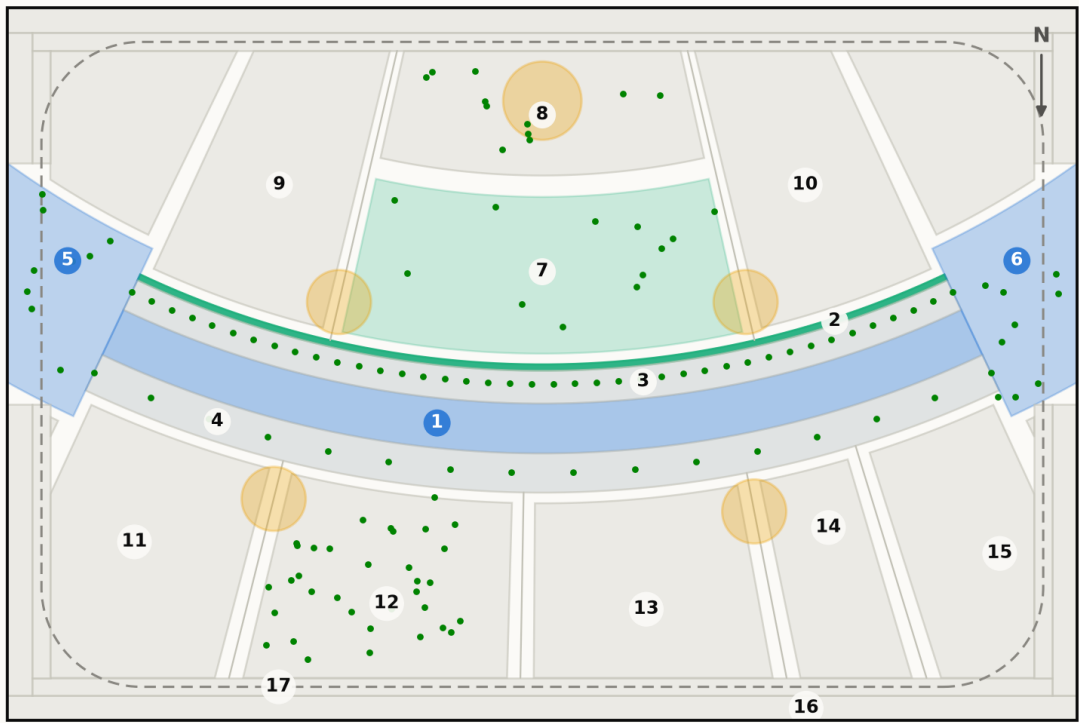
Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with python `run_analysis.py`

Fig10 masterplan

The plan as the code draws it — every room struck off the arc centre, areas measured from the drawn polygon

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from `src/plan.py`, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m ²
2	Al Falaj — the water channel	105 m ²
3	Crescent Shade Margin (N)	549 m ²
4	Crescent Shade Margin (S)	715 m ²
5	West Gate Majlis	439 m ²
6	East Gate Majlis	439 m ²
7	Al Nakhil — the Oasis Basin	1,140 m ²
8	Quiet Contemplation Garden	707 m ²
9	Children's Dune Play	1,267 m ²
10	Family Picnic Grove	1,267 m ²
11	Outdoor Fitness Terrace	908 m ²
12	Native Planting / Biodiversity Wadi	894 m ²
13	Community Plaza & Event Lawn	787 m ²
14	Souk Kiosks & Services	417 m ²
15	Multipurpose Sports Lawn	630 m ²
16	Al Kathib — the dune berm	1,464 m ²
17	Al Madar — the perimeter loop	986 m ²
	Al Sikkak — shaded alleys & setbacks	1,414 m ²
Total		15,000 m ²

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: `src/plan.py` — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

DRAWN BY [src/plan.py](#) [src/figures.py](#) **READ FROM** [data/raw/site_zoning_schedule.csv](#) [data/processed/masterplan_geometry.json](#)

Every path above is a live link. Nothing on this sheet was drawn by hand — regenerate it with python `run_analysis.py`

Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

PHASE 4 Concept Development

THIS DOCUMENT RESTS ON

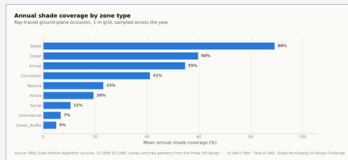


fig03_shade_by_zone.png

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE [src/plan.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/masterplan_geometry.json](#)

PHASE 5 Masterplan Development

THIS DOCUMENT RESTS ON

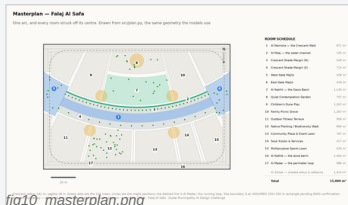


fig10_masterplan.png

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

CODE [src/plan.py](#) [src/figures.py](#)

PROOF — click to open [data/raw/site_zoning_schedule.csv](#)
[data/processed/masterplan_geometry.json](#)

PHASE 1 Site & Context Analysis

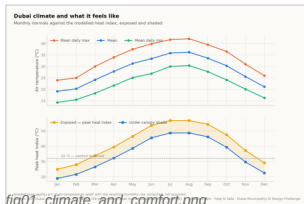


fig01_climate_and_comfort.png

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE [src/climate.py](#) [src/solar.py](#)

PROOF — click to open [data/processed/hourly_climate_comfort_8760.csv](#) [data/raw/sources.json](#)

PHASE 2 Problem Definition

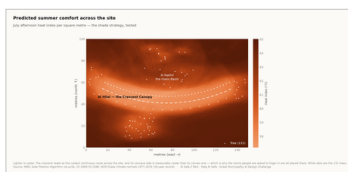


fig04_site_comfort_map.png

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE [src/climate.py](#)

PROOF — click to open [models/headline_metrics.json](#)

PHASE 3 Opportunity & Objectives

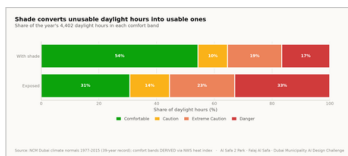


fig02_comfort_bands.png

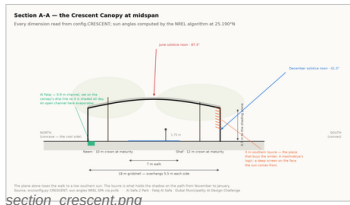
The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE [src/config.py](#) [src/plan.py](#)

PROOF — click to open [models/headline_metrics.json](#)

The work, with its proof attached — continued

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.



section_crescent.png

PHASE 6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

CODE [src/drawings.py](#) [src/config.py](#)

PROOF — [click to open](#) [data/processed/planting_layout.csv](#)

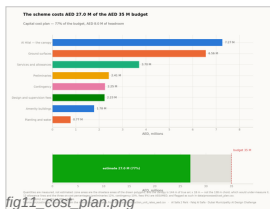


fig11_cost_plan.png

PHASE 7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

CODE [src/costing.py](#) [src/climate.py](#)

PROOF — [click to open](#) [data/processed/cost_plan.csv](#) [models/cost_summary.json](#)

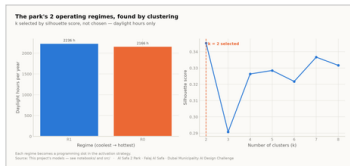


fig08_microclimate_regimes.png

PHASE 8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE [src/models.py](#) [src/dataset.py](#)

PROOF — [click to open](#) [data/processed/spatial_grid_comfort.csv](#)

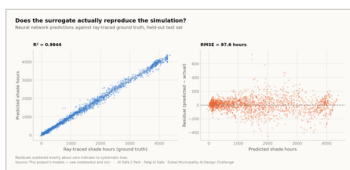


fig05_surrogate_performance.png

PHASE 9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

CODE [src/models.py](#) [src/boards.py](#) [tools/sync_film.py](#)

PROOF — [click to open](#) [models/model_metrics.json](#) [tests/test_pipeline.py](#)



board_2_evidence.png

PHASE 10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

CODE [tools/build_submission_pdfs.py](#) [tools/build_reports.py](#)

PROOF — [click to open](#) [tests/test_pipeline.py](#)

The 4 images in this file, and the whole record

This slot's own pictures come first, larger. The remaining 20 of the 24 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

01

IN THIS FILE

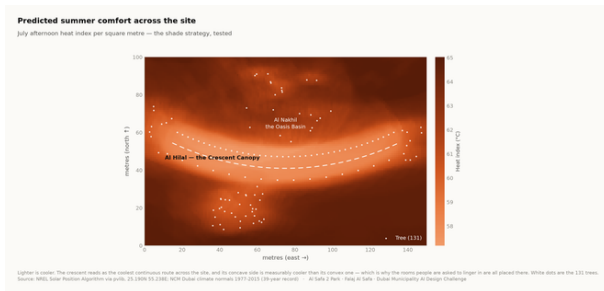


Fig04 site comfort map
analysis output

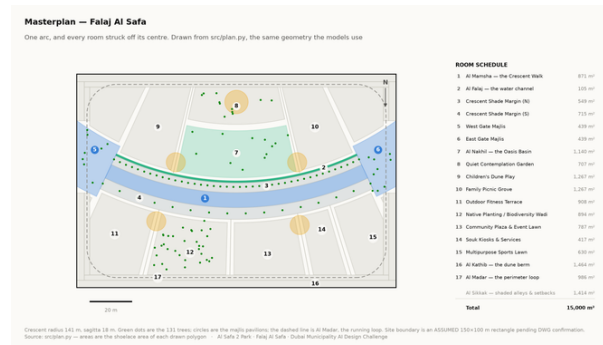


Fig10 masterplan
analysis output



Circulation
technical drawing



Facilities
technical drawing

02

EVERY OTHER IMAGE IN THE PROJECT

The other 20 images the project produces are not reprinted here — this file shows its own. Each name below opens the full-resolution original in the repository.

Fig01 climate and comfort analysis output

Fig05 surrogate performance analysis output

Fig08 microclimate regimes analysis output

Elevation technical drawing

Board 1 concept presentation board

Oasis basin artistic impression

Childrens dune play artistic impression

Fig02 comfort bands analysis output

Fig06 feature importance analysis output

Fig09 diurnal comfort analysis output

Planting technical drawing

Board 2 evidence presentation board

Spine corridor interior artistic impression

Souk plaza artistic impression

Fig03 shade by zone analysis output

Fig07 confusion matrix analysis output

Fig11 cost plan analysis output

Section technical drawing

Masterplan aerial golden hour artistic impression

Night plaza render artistic impression

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

144 m

CRESCENT ARC LENGTH

15,000 m²

SITE AREA

34.1%

SITE-WIDE MEAN SHADE

131

TREES PLANTED

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	4,402
Comfortable daylight hours — today	44.5%
Comfortable daylight hours — as designed	64.6%
Mean heat-index reduction under canopy	7.13 °C
Peak heat index, exposed	56.8 °C
Peak heat index, shaded	48.7 °C
Crescent Walk shaded, canopy and louvre	87.3%
Site-wide mean shade	34.1%
Trees planted	131

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R ²	0.9982
M1b Neural network — deployed surrogate, test R ²	0.9944
M2 Gradient Boosting — comfort band, test accuracy	97.5%
M3 K-Means — regimes selected by silhouette	k = 2
Training / validation / test split	10,500 / 2,250 / 2,250
Random seed, fixed for reproducibility	42

WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

Climate fidelity

the modelled year reproduces the published normals it was built from

No overlaps

no room in the schedule overlaps another

Areas close

every area sums back to 15,000 m²

Budget held

the cost plan stays inside AED 35 M

No leakage

no model can see its own answer in its inputs

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js   # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The complete project, and where to check it

Every one of the 125 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

THE SOURCES BEHIND THIS FILE

open these first if you want to check this document

[src/plan.py](#)

Technical drawing — to scale.

[src/drawings.py](#)

Technical drawing — to scale.

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Ma...
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Di...
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (6)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data

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[figures/fig06_feature_importance.png](#)
[figures/fig07_confusion_matrix.png](#)
[figures/fig08_microclimate_regimes.png](#)
[figures/fig09_diurnal_comfort.png](#)
[figures/fig10_masterplan.png](#)
[figures/fig11_cost_plan.png](#)

Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data
Analysis output — computed from project data

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (17)

[tools/__init__.py](#)
[tools/build_concept_film_hero.py](#)
[tools/build_concept_film_v2.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/check_renders.py](#)
[tools/embed_narration.py](#)
[tools/make_video_mp4.py](#)
[tools/make_winning_video.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)

Source dataset
Source dataset
Source dataset

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The complete project — continued

03

OF 12

data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	41 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file

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The complete project — continued

[submission/09_Material_Landscape_Palette/MANIFEST.md](#)
[submission/10_Complete_Design_Report/MANIFEST.md](#)
[submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md](#)
[submission/12_Concept_Animation_Video/MANIFEST.md](#)

What is in the slot, and what produced each file
What is in the slot, and what produced each file
What is in the slot, and what produced each file
What is in the slot, and what produced each file

WORKING HISTORY (1)

[archive/withdrawn_visuals/README.md](#)

Images withdrawn on purpose, and why